Working draft: learning results and abstract pending. This reading copy is not an ICRA template or a submission-compliance check.

Motion2Scene: Constructing Training Scenes from Executed Humanoid Motion Contrasts

Working manuscript, September 7, 2026. Abstract intentionally deferred until the registered result decision. This draft separates completed measurements from the unfinished M2S-ICRA-v1 comparison. It is not a submission-ready paper.

I. Introduction

A humanoid that can walk and crouch still needs to decide when crouching helps. Training this decision requires encounters whose observations distinguish the outcomes of the available commands. A randomly placed obstacle can be irrelevant to both commands, or defeat both. A beam placed above a crouching reference can appear useful while colliding with the robot during its actual transition into the crouch. The resulting label concerns an intended posture rather than the command the robot can execute from its current state.

We study scene construction from a pair of supported humanoid commands: commit to walking, or request a crouching reference at a fixed decision time. Both commands run through the same frozen controller. Their achieved trajectories provide a proposal model for placing overhead constraints, and paired obstacle-present simulations provide outcome labels. A fixed perceptive selector then predicts the success of each command. This separates three questions: whether a scene has a geometric contrast, whether that contrast survives execution, and whether the resulting data teaches a useful decision on other layouts.

Learning environments from motion is not itself our novelty claim. Learning from Learned Hallucination (LfLH) already learns obstacle configurations from open-space motion and uses hallucinated environments to train navigation policies [1]. Our controlled instance addresses a different supervision problem: full-body humanoid commands must enter and leave an adaptation through a legal transition, and an obstacle may change contact outcomes without making progress physically impossible.

The study makes three contributions at distinct evidence levels. First, it defines a command-bound construction and labeling protocol that preserves all four paired outcomes, including encounters that neither command solves. Second, it measures the mismatch between complete-reference screening and achieved-command screening, and validates source-specific switching banks and physical contrasts in Isaac Lab. Third, it specifies a four-arm comparison—uniform, analytic, target-only and learned construction—with one shared outcome learner. The third contribution remains an experimental question until the registered acquisition and evaluation complete. The learned proposal is a tested factor, not an assumed source of superiority.

II. Related work

LfLH learns hallucinated environments in which open-space plans are useful and trains reactive navigation from those environments [1]. This establishes the motion-to-scene- to-policy loop that motivates our experiment. We investigate how paired executed humanoid commands constrain the labels needed by that loop, rather than claiming that inverse scene generation is new.

Recent humanoid systems address perception-conditioned traversal and skill composition at broader scales. HumanoidPF represents humanoid–obstacle relationships for cluttered traversal [2]. Perceptive Humanoid Parkour composes atomic skills with motion matching and studies long-horizon vision-based execution [3]. Our experiment retains a narrow beam family and command bank to isolate training-scene construction. It does not compare against those systems' full task breadth or sensing pipelines.

SONIC supplies the inherited motion-tracking foundation and command interface [4]. We freeze its parameters and do not attribute its locomotion capacity to scene construction. The learning component is a supervised two-head outcome predictor. Because its command is issued once from a shared approach, the present task is not an end-to-end reinforcement-learning study. Sequential extensions would need to address policy-induced state distributions, as emphasized by DAgger [5]. Evaluation reports paired outcomes and the small number of carrier groups, consistent with the need to expose uncertainty rather than infer robustness from point estimates [6].

III. Problem and command-bound supervision

Let S denote a static scene, x_t the measured robot state, phi_t the reference phase, h_t the relevant approach history, and a a supported command. Define the binary task outcome as Y_a = Y(S, x_t, phi_t, h_t, a). A0 commits to walking through the encounter. A1 requests d040 at t=0.30 s and uses the shared legal return interface. A0 is not a wait-and-reconsider action. A refusal from the predictor continues A0; it is not a physical stop or successful avoidance.

The contact-qualified task requires at most 1 N measured beam force through passage, body-origin crossing 0.1 m beyond the beam, and 0.3 s upright stability in the first episode. We record resets, falls, legal entry/return and endpoint tracking separately. This definition admits contact-qualified passage with an endpoint-tracking rejection; it does not imply exact reference completion or an all-time collision guarantee.

Each encounter receives the pair (Y_0,Y_1). Both-pass examples permit a preference for walking. Walk-fail/crouch-pass examples establish useful adaptation. Walk-pass/crouch-fail examples warn against the transition. Both-fail examples describe the limits of the current command bank. Missing or mismatched executions are masked rather than converted to negative labels. Paired executions must agree on recorded pre-decision state, sensing, action and token histories. This is measured prefix agreement, not a claim to restore an unobserved complete controller state.

IV. Construction and fixed learner

For each qualified carrier, empty-scene executions produce achieved walk and d040 body trajectories. A capsule representation supplies inexpensive geometric queries. The station coordinate follows the source reference route; beam dimensions are 0.10 by 1.20 by 0.10 m, with station in [0.1,0.9] and underside in [1.1,1.45] m. The achieved forecast includes entry, passage and exit. Physics remains the labeling oracle because obstacle interaction and tracking deviations can invalidate that forecast.

Uniform construction samples the common domain without a future-trajectory contrast filter. Analytic construction uses achieved envelopes and global distinct search. Motion2Scene retains the frozen local-event initializer and 17-query bounded pattern search, replacing its geometric evaluator with achieved trajectories. The target-only ablation uses its previously fitted no-contrast model and removes walking interference from search and acceptance. All arms share finite-domain, duplicate, reserved-layout and pre-decision validity checks. Target and contrast screens retain the declared perturbation offsets; every rejected attempt remains charged.

The observation contains 144 values from twelve upper rays and their conditional lower-ray queries, plus 70 state/phase/history values. Neither source identity nor true beam coordinates enters a learned selector. One deterministic logistic model predicts two probabilities from the 214 inputs. All inputs have physical scale one except joint velocity, scaled by five. Zero initialization, 2000 Adam updates at 0.01 and a 0.01 mean-squared-weight penalty are common across arms. The >=0.5 rule prefers walking when both heads are positive and requests d040 only when walking is predicted infeasible and d040 feasible. The transition manager enforces legality; it does not make a hidden scene-dependent choice beneath the model.

V. Completed physical and mechanism evidence

The earlier generated-source pilot admits six source pairs from eight requested and obtains eighteen contact-avoidance contrasts from twenty-four requested scene slots. Walking still crosses in those cases but violates the contact criterion. Thus the result distinguishes task outcomes, not physical impossibility. d040 also passes the selected 41002 beam in three runs, so d055 is not established as minimally necessary. [Internal evidence: SOURCE_EXECUTION_V1_RESULT, BEAM_D040_EXECUTION_V1_RESULT.]

The selector-breakpoint study identifies one d040 rescue among six matched layout/seed conditions on carrier 41002. A shared analytic-trained linear control realizes this rescue: at underside 1.27 m and seed 8512, d040 records 0 N through qualifying passage, where matched walking records 57.947 N. Its 4/6 passage contrasts with 3/6 for the other three data arms. This is development evidence on one inspected carrier. Changing scaling, capacity and regularization together does not isolate normalization as the sole causal explanation of the original MLPs' failure.

The achieved-transition forecast catches thirteen command contacts that the complete- reference screen falsely predicts clear, but misses two other contact-qualified failures. It improves a measured diagnostic and remains imperfect. The next empty- command bank qualifies 41001, 41002 and 41003 in both seeds: twelve executions, six paired approaches and legal d040 entry/return. Qualification does not require every reference-tracking metric to pass.

The first bounded construction pilot accepts only one of twelve assigned slots, from analytic construction on 41001; no learned slot survives. Both seeds then show walking contact (1635.077 and 1130.936 N) and d040 contact-qualified passage (0 N). Both d040 runs also receive reference_endpoint_tracking_error rejections. The full funnel therefore demonstrates one unique analytic scene, not broad learned-scene yield. A finite support map finds 67 nominal candidate centres after other checks, but only one under the inherited perturbation audit. None of its four pre-filter robust centres lies inside the frozen learned draws' trust boxes. These finite samples do not prove infeasibility or justify revising the earlier outcomes.

VI. Registered learning-utility experiment

M2S-ICRA-v1 assigns 24 encounter groups per arm across the three qualified development carriers: eighteen generated and six identical shared backgrounds. All four methods retain their refused assignments. At most 156 unique command executions label the four arms and shared backgrounds. If acquisition produces fewer complete groups, we report actual counts and treat the result as an equal-requested-slot comparison; we cannot claim to have completed an equal-24-label experiment.

Four fixed learners and two declared comparators face twelve reserved station/height layouts, two physics seeds and three carriers, yielding 432 traversal assignments. The comparators are the upper/lower-ray decision rule and a privileged achieved- geometry predictor, not an optimal transition-time oracle. Passage is averaged within carrier before averaging carriers. Paired tables report both-pass, method-A-only, method-B-only and both-fail. Repeats share carriers and do not create independent source evidence. Background adaptation/refusal outcomes form separate suites.

[Results pending: complete acquisition funnel, paired labels, fitted controls, 432-cell traversal panel, comparator characterization and requested-slot cost table. Do not replace these entries with predictions or offline command lookups.]

VII. Limitations and result decision

This is a controlled instance with one beam family, two commands, ideal simulator rays, three inspected development carriers and one frozen controller. It has no hardware or source-held-out transfer claim. The original 120/600 MLP evaluation remains a retained partial failure, with 480 assignments paused. Existing layout diagnostics have informed this revision, so their reuse is development evaluation.

Geometric proposal acceptance does not guarantee a useful physical contrast. The current learned initializer has produced no accepted scenes in the achieved-transition pilot, and the matched 24-label quota may fail. Contact-qualified passage does not eliminate endpoint tracking error. Neither a predicted refusal nor continued neutral walking is a qualified protective action. The final conclusion must follow the registered comparison: report contrast-construction utility if supported, learned proposal benefit only if measured, and a construction/validation result if learning utility remains unresolved.

References (verified primary records, September 7)

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